



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

20-01-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	C	C	A	B	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	C	C	A	B	A	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	8	8	2	4	2	5	5	1	1

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	B	D	C	D	C	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	B	B	C	A	D	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	10	70	6	25	6	9	8	6	1	11

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	A	B	A	C	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	B	B	B	B	C	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	1010	9	4	2	4	2	14	6	3

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (D)

$$S = \alpha^2 \beta \ln \left(\frac{\mu K R}{J \beta^2} + 3 \right)$$

$$S = \frac{Q}{T} = \text{joule/k}$$

$$[\alpha^2 \beta] = \text{Joule/k}$$

$$PV = nRT \quad \left[\frac{\mu K R}{J \beta^2} \right] = 1$$

$$R = \frac{\text{Joule}}{K}$$

$$\Rightarrow R = \frac{\text{Joule}}{K}, K = \frac{\text{Joule}}{R}$$

$$\Rightarrow \beta = \left(\frac{\text{Joule}}{K} \right)$$

$$\alpha^2 \beta = \left(\frac{\text{Joule}}{K} \right)$$

$$\Rightarrow \alpha = \text{dimensionless}$$

2. Ans (D)

Truth table of the given gate :

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Truth table of option (1)

A	B	C
0	0	1
0	1	1
1	0	0
1	1	1

Truth table of option (2)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	1

Truth table of option (3)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Truth table of option (4)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Since option (1) has same truth table, hence answer is option (4) is correct.

Alternative solution :

Given Boolean expression can be written as

$$A + \bar{B} = C$$

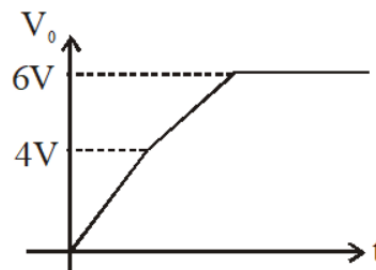
$$\therefore C = \bar{A} \cdot \bar{B} = \bar{A} \cdot B$$

Hence option (D) is correct

3. Ans (D)

Till input voltage Reaches 4V No zener is in Breakdown Region So $V_0 = V_i$ Then Now when V_i changes between 4V to 6V One Zener with 4V will Breakdown are P.D. across This zener will become constant and Remaining Potential will drop. acro Resistance in series with 4V Zener.

Now current in circuit increases Abruptly and source must have an internal resistance due to which. Some potential will get drop across the source also so correct graph between V_0 and t . Will be



We have to Assume some resistance in series with source.

4. Ans (C)

$$I_T = I_C \sqrt{1 + \frac{m^2}{2}}$$

$$\Rightarrow I_C = \frac{11}{\sqrt{1 + \frac{(0.5)^2}{2}}} = 10.3$$

5. Ans (C)

Given 10 VSD = 9 MSD

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

Here MSD → Main Scale division

VSD → Vernier Scale division

Least count = 1 MSD – 1 VSD

$$= \left(1 - \frac{9}{10}\right) \text{ MSD} = 0.1 \text{ MSD}$$

$$= 0.1 \times 0.1 \text{ cm} = 0.01 \text{ cm}$$

As '0' of V.S. lie before '0' of M.S.

Zero error = $-[10 - 6] \text{ L.C.}$

$$= -4 \times 0.01 \text{ cm} = -0.04 \text{ cm}$$

$$\text{Reading} = 3.1 \text{ cm} + 1 \times \text{LC}$$

$$= 3.4 \text{ cm} + 1 \times 0.01 \text{ cm} = 3.11 \text{ cm}$$

True diameter = Reading – Zero error

$$= 3.11 - (-0.04) \text{ cm} = 3.15 \text{ cm}$$

6. **Ans (A)**

$$\frac{T_L}{T_H - T_L} = \text{C.O.P.} = \frac{\frac{dH}{dt}}{\frac{dW}{dt}}$$

$$\frac{263}{35} \times 35 = \frac{dH}{dt}$$

$$\frac{dH}{dt} = 263 \text{ watts}$$

Ans.1

9. **Ans (B)**

$$I = \epsilon_0 C E_{\text{rms}}^2$$

$$E_{\text{rms}} = c B_{\text{rms}}$$

$$I = \epsilon_0 C^3 B_{\text{rms}}^2$$

$$B_{\text{rms}} = \sqrt{\frac{I}{\epsilon_0 C^3}}$$

$$B_{\text{rms}} \approx 10^{-4}$$

11. **Ans (B)**

Polaroid

$$I_0 \left| \frac{I_0}{2} \right| I' = \frac{I_0}{2} \cos^2 30^\circ |I'' = I' \cos^2 60^\circ$$

$$I' = \frac{3I_0}{8} \quad I'' = \frac{3I_0}{32}$$

$$\frac{I''}{I_0} \times 100 = 94\%$$

12. **Ans (A)**

$$U = -\frac{C}{r}$$

$$F = -\frac{dU}{dr} = -\frac{C}{r^2}$$

$$|F| = \frac{mv^2}{r}$$

$$\frac{C}{r^2} = \frac{mv^2}{r}$$

$$v^2 \propto \frac{1}{r}$$

14. **Ans (C)**

$$\lambda_B = 2\lambda_A$$

$$\Rightarrow \frac{h}{\sqrt{2T_B m}} = \frac{2h}{\sqrt{2T_A m}}$$

$$T_A = 4T_B \quad \dots(i)$$

$$\text{and } T_B = (T_A - 1.5) \text{ eV} \quad \dots(ii)$$

from (i) and (ii)

$$3T_B = 1.5 \text{ eV} \Rightarrow T_B = 0.5 \text{ eV}$$

$$T_B = 0.5 \text{ eV} = 4.5 \text{ eV} - \phi_B$$

$$\phi = 4 \text{ eV}$$

15. **Ans (C)**

$$\sqrt{v} \propto (z-1) \quad [\text{For } k_\alpha \text{ lines}]$$

$$v \propto (z-1)^2$$

$$\frac{1}{\lambda} \propto (z-1)^2$$

$$\frac{\lambda_{43}}{\lambda_{29}} = \frac{(29-1)^2}{(43-1)^2}$$

$$\lambda_{29} = \frac{9}{4} \lambda$$

16. **Ans (A)**

$$\omega = \frac{\text{angular momentum}}{mR^2 (\text{M.I.})} = \frac{J}{mR^2}$$

Ring is equivalent to a current loop of current

$$I = \frac{Q}{2\pi} \omega$$

magnetic moment of loop

$$\vec{\mu} = nI\vec{A} = (1) \left(\frac{Q}{2\pi} \omega \right) \pi R^2 \hat{n} = \left(\frac{Q}{2} \omega R^2 \right) \hat{n}$$

($\hat{n}$ is normal to plane outward $\odot$)

$$\text{Torque } \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$= \frac{Q}{2} \omega R^2 B \text{ (up the plane)}$$

$$= \frac{Q}{2} R^2 \left(\frac{J}{mR^2} \right) B = \frac{1}{2} \left(\frac{Q}{m} \right) (J) (B)$$

$$= \frac{1}{2} \times 10^{-5} \times 7.5 \times 10^{-4} \times 0.1$$

$$= 3.75 \times 10^{-10} \text{ N upward along the plane}$$

18. **Ans (A)**

$$v_y = \sqrt{\frac{T_y}{\mu_y}}$$

$$T_y = \left\{ \int_0^y \mu_0 e^y dy \right\} g$$

$$T_y = \mu_0 (e^y - 1) \cdot g$$

$$v_y = \sqrt{g - \frac{g}{e^y}}$$

$$v_y^2 = g (1 - e^{-y})$$

19. **Ans (B)**

$$A = A_0 e^{-\left(\frac{b}{2m}\right)t}$$

$$\frac{A_0}{2} = A_0 e^{-\frac{b}{2m}(1 \text{ min})}$$

$$\frac{1}{2} = e^{-\frac{b}{2m}(1 \text{ min})}$$

$$-\frac{b}{2m}(1 \text{ min}) = -\ln 2 \dots (1)$$

$$\text{Now } A = A_0 e^{-\frac{b}{2m}(3 \text{ min})}$$

$$A = A_0 e^{-3 \ln 2}$$

$$\therefore A = 2^{-3} A_0$$

20. **Ans (C)**

$$\vec{B} = \frac{\mu_0 I}{4\pi L} \sin 45^\circ (-\hat{k}) + \frac{\mu_0 I \pi}{4\pi \frac{L}{2}} (-\hat{k}) + \frac{\mu_0 I}{4\pi \frac{L}{4}} \times \frac{\pi}{2} (-\hat{k})$$

PART-1 : PHYSICS

SECTION-II

2. **Ans (8)**

Loss in P.E. = Gain in K.E.

$$mg \frac{\ell}{3} + mg \left(\frac{2\ell}{3} \right) + mg \ell$$

$$= \frac{1}{2} \left(m \left(\frac{\ell}{3} \right)^2 + m \left(\frac{2\ell}{3} \right)^2 + m \ell^2 \right) \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{36g}{14\ell}} \Rightarrow v_B = \omega \ell_B = \frac{2\ell}{3} \sqrt{\frac{36g}{14\ell}} = \sqrt{\frac{8g\ell}{7}}$$

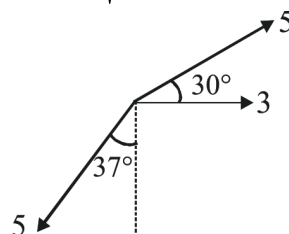
$$= 8$$

4. **Ans (2)**

$$v_1 = 3 \sin \omega t; v_2 = 5 \sin(\omega t + \phi_1);$$

$$v_3 = 5 \sin(\omega t - \phi_2)$$

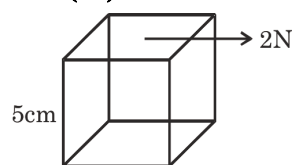
$$v_{\max} = \sqrt{\left(\frac{5\sqrt{3}}{2} \right)^2 + (1.5)^2} = \sqrt{21}$$



$$\frac{I_{\max}}{R} = \frac{V_{\max}}{R} = \frac{\sqrt{21}}{\sqrt{7/3}} = \sqrt{\frac{21 \times 3}{7}} = 3A$$

$$I_{\text{rms}} = \frac{3}{1.4} = 2.14$$

5. **Ans (4)**



$$\text{Shear strain} = \frac{1 \times 10^{-3}}{5 \times 10^{-2}} = \frac{1}{50}$$

$$\eta = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F/A}{\text{Shear strain}}$$

$$= \frac{2}{25 \times 10^{-4}} \times 50 = 4 \times 10^4 \Rightarrow x = 4$$

6. **Ans (2)**

Net loss = k (temperature different between both sphere & surrounding)

$$ms \frac{d\theta}{dt} = -k[\theta - \theta_s]$$

$$\int_{\theta_2=0}^{\theta_2=\frac{s}{2}} \frac{Cd\theta}{(\theta - \theta_s)} = \int_0^t k dt$$

$$= -C[\ln(\theta - \theta_s)]_0^{\frac{s}{2}} = kt$$

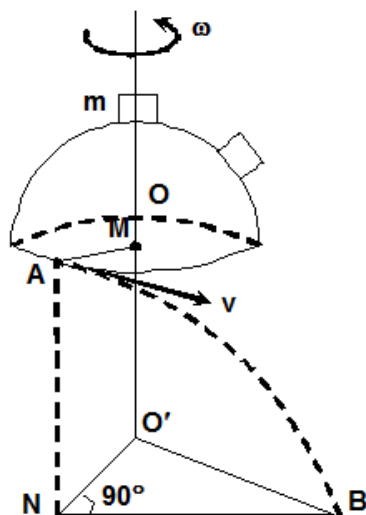
$$-C \ln \left(\frac{-\theta_s}{-\theta_s} \right) = kt$$

$$-C \ln \left(\frac{1}{2} \right) = kt$$

$$C \ln(2) = kt$$

$$t = \frac{C}{k} \ln(2)$$

7. Ans (5)



Let the drop is thrown off from point A. The

$$\text{velocity } V = r\omega = \frac{3}{2} \times \frac{28\pi}{33} = 4 \text{ ft s}^{-1}$$

The drop will describe the parabolic path

The time take by drop to strike the ground $\frac{1}{2}$

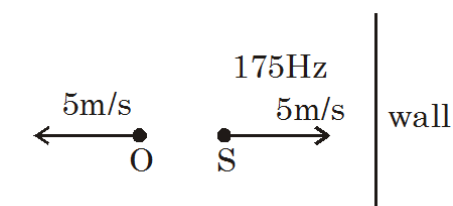
$$gt^2 = 4 \Rightarrow t = \frac{1}{2} \text{ s}$$

$\therefore$ The horizontal distance $NB = 4 \times \frac{1}{2} \text{ ft} = 2 \text{ ft}$

$$\therefore BO' = \sqrt{ON^2 + NB^2} = \sqrt{\left(\frac{3}{2}\right)^2 + (2)^2} = \frac{5}{2} \text{ ft}$$

i.e., diameter = 5 ft.

8. Ans (5)

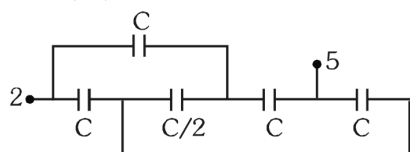


$$f_1 = \left(\frac{355 - 5}{355 + 5} \right) 175$$

$$f_2 = \left(\frac{355 - 5}{355 - 5} \right) \times 175 \times \left(\frac{355 - 5}{355} \right)$$

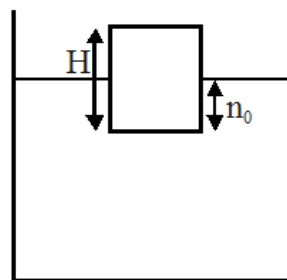
$$f_{\text{beat}} = |f_1 - f_2| = 4.86$$

9. Ans (1)



Redraw the circuit $C_{\text{eq}} = C = \frac{\epsilon_0 A}{d}$

10. Ans (1)



$$\rho_0 \times A \times n_0 \times g = \rho_0 \times A \times g \times H$$

$$\Rightarrow x_0 = \frac{H \times \rho_i}{\rho_0}$$

$$\omega_E + \omega_g + \omega_B = 0$$

$$\omega_E + Mg \times (H - x_0) + \left[\int_{x_0}^H \rho_0 x A g \, dx \right] = 0$$

$$\Rightarrow \omega = \frac{Agh^2(\rho_0 - \rho_i)^2}{2\rho_0}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

The colour of ruby is due to presence of Cr(III) ions, randomly embedded into Alumina lattice in place of some Al^{3+} ions.

3. Ans (B)

CFC's cause stratospheric pollution.

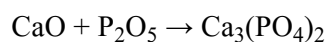
Dust particle are fine solids having diameter more than 1 mm.

Classical smog doesn't occur in warm and dry conditions.

4. Ans (B)

Water gas shift reaction consumes syngas.

5. Ans (D)

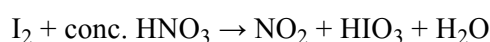


This reaction occurs in bessemerization

6. Ans (C)

Hydrolysis of borax gives NaOH and H_3BO_3 .

7. Ans (D)



6. Ans (9)

$$\psi = \left(\frac{2}{\pi C^2} \right)^{\frac{3}{4}} \frac{e^{-(x^2+y^2+z^2)}}{C^2}$$

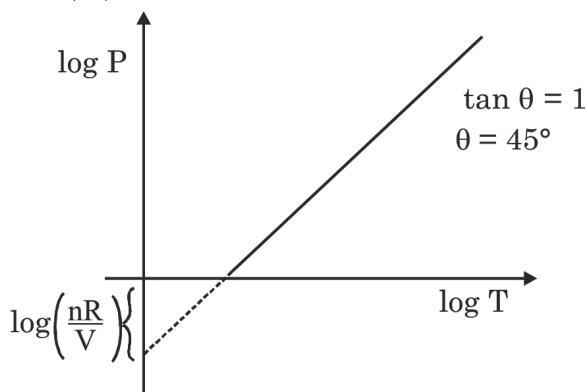
$$|\psi|^2 dx dy dz = \left(\frac{2}{\pi C^2} \right) \frac{e^{-2(x^2+y^2+z^2)}}{C^2} dx dy dz$$

$$= \left[\frac{2}{(4\pi n m^2)} \right]^{3/2} \frac{e^{-2(1.2)^2 + (-1)^2 + (0)^2}}{4} (0.004 nm)^3$$

$$= 1.200 \times 10^{-9}$$

Ans. 9

9. Ans (1)



$$PV = nRT$$

$$V = \text{constant}$$

$$P = \frac{nR}{V} T$$

$$\log P = \log \frac{nR}{V} + \log T$$

$$\text{Slope (a)} = 1$$

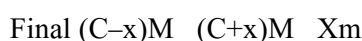
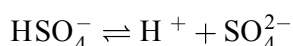
$$\text{y-intercept (c)} = \log \left(\frac{nR}{V} \right)$$

$$= \log \left(\frac{10 \times 0.0821}{82.1} \right) = -2$$

$$\text{x-intercept (b)} = +2 \quad \therefore \theta = 45$$

$$\Rightarrow a + b + c = 1 + 2 - 2 = 1$$

10. Ans (11)



$$\text{Now, } 1.2 \times 10^{-2} = \frac{0.01 \times x}{C - x} \Rightarrow x = \frac{6}{11} C$$

$$\text{And } C + x = 0.01 \Rightarrow C = \frac{0.01 \times 11}{17} M$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\tan^{-1} \left(\frac{x + \frac{2}{x} - x + \frac{2}{x}}{1 + x^2 - \frac{4}{x^2}} \right)$$

$$\therefore \frac{\frac{4}{x}}{1 + x^2 - \frac{4}{x^2}} = \frac{4}{x}$$

$$1 + x^2 - \frac{4}{x^2} = 1$$

$$x = \pm \sqrt{2}$$

$x = 4$ not satisfying.

2. Ans (B)

Property, SP = Geometric mean of length of tangents

3. Ans (D)

$$\sum_{r=0}^{10} (-1)^r 10C_r \left(\frac{1}{3} \right)^r + \sum_{r=0}^{10} (-1)^r 10C_r \left(\frac{8}{9} \right)^r$$

$$\left(1 - \frac{1}{3} \right)^{10} + \left(1 - \frac{8}{9} \right)^{10}$$

$$\frac{2^{10}}{3^{10}} + \frac{1}{9^{10}}$$

$$\frac{6^{10} + 1}{3^{20}}$$

4. Ans (A)

$$\vec{c} = \lambda \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right)$$

5. Ans (B)

$$T_n = \frac{1}{(n+1)(n+2)(n+3)} + \frac{1}{n(n+1)(n+2)(n+3)}$$

$$T_n = \frac{1}{2} \left[\frac{1}{(n+1)(n+2)} - \frac{1}{(n+2)(n+3)} \right]$$

$$\frac{1}{3} \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right]$$

$$\therefore S_{\infty} = \frac{5}{36}$$

6. Ans (A)

$$S = 1 + 2 \left(\frac{10}{11} \right) + 3 \cdot \left(\frac{10}{11} \right)^2 + \dots \infty$$

$$\text{(Let } \frac{10}{11} = x)$$

$$S = 1 + 2x + 3x^2 + \dots \infty$$

$$S \cdot x = x + x^2 + \dots \infty$$

$$S(1-x) = 1 + x + 2x^2 + \dots \infty$$

$$S(1-x) = \frac{1}{1-x}$$

$$S = \frac{1}{(1-x)^2} = \frac{1}{\left(1 - \frac{10}{11}\right)^2} = 121$$

7. Ans (C)

$$\begin{aligned} \int_0^2 f(x) dx &= \int_0^1 f(x) dx + \int_1^2 f(x) dx = \int_0^1 (f(x) + f(x+1)) dx \\ &= \int_0^1 (x^2 + 2x + 3) dx = \frac{x^3}{3} + x^2 + 3x = \frac{1}{3} + 1 + 3 = \frac{13}{3} \end{aligned}$$

8. Ans (C)

$$f(x) = \frac{e^x - e^{-x}}{10e^x + e^x} = 0 \text{ for } x \geq 0$$

$$= \frac{e^x - e^{-x}}{10e^x + e^{-x}} = \frac{e^{2x} - 1}{10e^{2x} + 1} \text{ for } x < 0$$

$$\text{for } x < 0, f(x) = \frac{e^{2x} - 1}{10(e^{2x} + \frac{1}{10})} = \frac{e^{2x} + \frac{1}{10} - \frac{11}{10}}{10(e^{2x} + \frac{1}{10})}$$

$$= \frac{1}{10} \left[1 - \frac{\frac{11}{10}}{e^{2x} + \frac{1}{10}} \right]$$

$$\frac{1}{10} < e^{2x} + \frac{1}{10} < \frac{1}{10} + 1$$

$$f(x) \in (-1, 0). \text{ Range of } f(x) \text{ is } (-1, 0]$$

9. Ans (B)

$$\text{Sum 4} = \{(1, 3), (2, 2), (3, 1)\}$$

$$\text{Sum 7} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

required probability

$$= \frac{1}{6} + \frac{27}{36} \cdot \frac{1}{6} + \left(\frac{27}{36} \right)^2 \cdot \frac{1}{6} + \dots \infty$$

$$= \frac{\frac{1}{6}}{1 - \frac{27}{36}}$$

10. Ans (B)

$$z^2 = \bar{z}, \Rightarrow \bar{z}^2 = z \Rightarrow z^4 = z \Rightarrow 4 \text{ solutions}$$

$$\text{So, coefficient of } x^{11} \text{ in } (2x^2 + x - 3)^6$$

$$= \text{coefficient of } x^{11} \text{ in } (x-1)^6 (2x+3)^6$$

$$= {}^6C_0 \times {}^6C_1 2^5 \times 3 - {}^6C_1 \times {}^6C_1 \times {}^6C_0 2^5$$

$$= 3 \times 6 \times 2^5 - 6 \times 2^6 = 6 \times 2^5 (3-2) = 192$$

or

the term having x^{11} by multinomial

$$\text{is } \frac{6!}{0!1!5!} (-3)^0 (x)^1 (2x^2)^5$$

11. Ans (C)

$$AB = A \Rightarrow A^2 = A, BA = B \Rightarrow B^2 = B$$

$$\Rightarrow (A+B)^{2020} = 2^{2019}A + 2^{2019}B = 2^{2019}(A+B)$$

12. Ans (B)

$$x^3 + 3x^2 - 3x - 5 = (x+1)(x^2 + 2x - 5)$$

$$x^3 + ax^2 + bx = x(x^2 + ax + b)$$

$$\text{Common root can be } -1 \Rightarrow a - b = 1$$

$$\text{or } a = 2, b = -5$$

13. Ans (D)

See from truth table

14. Ans (B)

$$\text{Let } P(x_1, y_1) \text{ on the curve } \frac{dy}{dx} = -\frac{x^3}{y^3} = -\frac{x_1^3}{y_1^3}$$

Equation of tangent at $P(x_1, y_1)$.

$$y - y_1 = -\frac{x_1^3}{y_1^3}(x - x_1), y_1^3 \cdot y - y_1^4 = -x \cdot x_1^3 + x_1^4,$$

$$y_1^3 \cdot y + x \cdot x_1^3 = x_1^4 + y_1^4 = c$$

$$\frac{y}{\frac{c}{y_1^3}} + \frac{x}{\frac{c}{x_1^3}} = 1, a = \frac{c}{x_1^3} \text{ and } b = \frac{c}{y_1^3}, x_1 = \left(\frac{c}{a} \right)^{\frac{1}{3}},$$

$$y_1 = \left(\frac{c}{b} \right)^{\frac{1}{3}}, x_1^4 + y_1^4 = c, \left(\frac{c}{a} \right)^{\frac{4}{3}} + \left(\frac{c}{b} \right)^{\frac{4}{3}} = c$$

$$a^{-\frac{4}{3}} + b^{-\frac{4}{3}} = c^{-\frac{1}{3}}$$

$$\Rightarrow \alpha = \frac{4}{3}, \beta = \frac{2}{3}, \gamma = \frac{1}{3}$$

$\Rightarrow \alpha, \beta, \gamma$ are in Geometric Progression.

15. Ans (B)

Let $\cos \theta = x$,

$$\sqrt{2x^2 - 4x + 6} + \sqrt{19 - 6x + 3x^2} = 5 - x^2 + 2x$$

$$\text{or } \sqrt{2(x^2 - 2x + 1) + 4} + \sqrt{3(x^2 - 2x + 1) + 16} = 6 - (x^2 - 2x + 1)$$

$$\text{or } \sqrt{2(x-1)^2 + 4} + \sqrt{3(x-1)^2 + 16} = 6 - (x-1)^2$$

L.H.S. ≥ 6 and R.H.S. ≤ 6 .

But L.H.S. = R.H.S.

Hence, $(x-1)^2 = 0$, $x = 1$,

$\cos \theta = 1$, $\theta = 2n\pi$, $\theta = -2\pi$, 0 , 2π

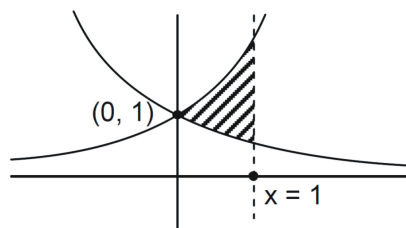
Number of solution = 3

16. Ans (B)

$$y^2 - 4x - 4y + 12 = 0$$

$$\Rightarrow y^2 - 4y + 4 = 4x - 8$$

$\Rightarrow (y-2)^2 = 4(x-2)$ directrix of the above curve is required area.



$$\begin{aligned} &= \int_0^1 (e^x - e^{-x}) dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - 2 \\ &= \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots\right) + \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots\right) - 2 \\ &= 2 \left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\right) = 2 \sum_{r=1}^{\infty} \frac{1}{(2r)!} \end{aligned}$$

17. Ans (B)

$$I_2 - I_1 = \int_1^3 (3x^2 - 4x - 5)f(x^3 - 2x^2 - 5x + 2020) dx.$$

$$\text{Put } x^3 - 2x^2 - 5x + 2020 = z$$

$$(3x^2 - 4x - 5)dx = dz,$$

$$\text{when } x = 1, 1 - 2 - 5 + 2020 = z_{\min},$$

$$z_{\min} = 2014$$

$$\text{When } x = 3, 27 - 2 \times 9 - 5 \times 3 + 2020 = z_{\max},$$

$$z_{\max} = 2014 \Rightarrow I_2 - I_1 = \int_{2014}^{2014} f(z) dz = 0$$

$$I_2 = I_1 \Rightarrow \frac{2I_1}{3I_2} = \frac{2}{3} \rightarrow (0, 1)$$

18. Ans (C)

The circles touch each other at $(1, 1)$

$$x(x-1) + y(y-1) + \lambda(x-y) = 0$$

$$x^2 + y^2 + x(\lambda-1) + y(-\lambda-1) = 0$$

radius = 1

$$\sqrt{\frac{(\lambda-1)^2}{4} + \frac{(\lambda+1)^2}{4}} = 1$$

$$\frac{1}{2}(\lambda^2 + 1) = 1$$

$$\lambda^2 + 1 = 2$$

$$\lambda = \pm 1$$

$\therefore$ two circles

19. Ans (C)

Since $\hat{a}$ and $\hat{b}$ are perpendicular hence

$\hat{a}$, $\hat{b}$ and $\hat{a} \times \hat{b}$ are non coplanar.

Hence any vector say $\vec{r}$ can be expressed as

$$\therefore \vec{r} \times \hat{a} = x\hat{a} + y\hat{b} + z(\hat{a} \times \hat{b}) \quad \dots(1)$$

$$\text{dot with } \vec{a} \quad 0 = x + 0 + 0 \Rightarrow x = 0$$

$$\text{dot with } \vec{b} \quad [\vec{r} \hat{a} \hat{b}] = 0 + y + 0$$

$$\Rightarrow y = [\vec{r} \hat{a} \hat{b}]$$

$$\text{dot with } \vec{r} \quad 0 = \underbrace{x\hat{a} \cdot \vec{r}}_{\text{zero as } x=0} + y\vec{r} \cdot \hat{b} + z[\vec{r} \hat{a} \hat{b}]$$

$$0 = [\vec{r} \hat{a} \hat{b}](\vec{r} \cdot \hat{b}) + z[\vec{r} \hat{a} \hat{b}]$$

$$\Rightarrow z = -(\vec{r} \cdot \hat{b})$$

$$\text{Hence } \vec{r} \times \vec{a} = [\vec{r} \hat{a} \hat{b}] \hat{b} - (\vec{r} \cdot \hat{b}) (\hat{a} \times \hat{b})$$

$$\vec{r} \times \hat{a} = [\vec{r} \hat{a} \hat{b}] \hat{b} + (\vec{r} \cdot \hat{b}) (\hat{b} \times \hat{a})$$

20. Ans (A)

$$2 \cos^{-1} x = \cot^{-1} \left(\frac{2x^2 - 1}{2x \sqrt{1-x^2}} \right)$$

$$\text{put } x = \cos \theta; \text{ LHS} = 2\theta;$$

$$0 \leq \theta \leq \pi \text{ and } -1 \leq x \leq 1 \quad \dots(1)$$

$$\text{R.H.S.} = \cot^{-1} \left(\frac{\cos 2\theta}{2 \cos \theta |\sin \theta|} \right)$$

$$= \cot^{-1}(\cot 2\theta) = 2\theta \text{ if } 0 < 2\theta < \pi \quad \dots(2)$$

$$\text{or } 0 < \theta < \pi/2$$

$$\text{from (1) and (2) } 0 < \theta < \pi/2$$

$$\therefore x \in (0, 1)$$

PART-3 : MATHEMATICS

SECTION-II

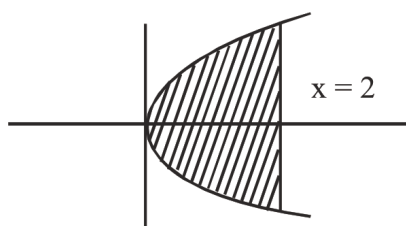
1. **Ans (5)**

let $p(x, y)$ be any point on the curve,

$$\text{Now } \frac{dy}{dx} = \frac{1}{y} \Rightarrow y \, dy = dx \Rightarrow \frac{y^2}{2} = x + c$$

$\therefore$ if passes through $M(2, 2)$ so $c = 0$

$$y^2 = 2x$$



$$\text{Here required area} = 2 \int_0^2 \sqrt{2x} \, dx$$

$$\begin{aligned} &= 2\sqrt{2} \left[\frac{2}{3} x^{3/2} \right]_0^2 \\ &= 2\sqrt{2} \cdot \frac{2}{3} \cdot 2\sqrt{2} = \frac{16}{3} \\ &= 5.33 \end{aligned}$$

2. **Ans (1010)**

$$z = e^{i \left(\frac{\ln \pi + \frac{\pi}{2}}{2021} \right)} \text{ where } n = 0$$

$$\operatorname{Re}(z) > 0 \text{ for } n \in [0, 504] \cup [1516, 2020] \quad (n \in I)$$

3. **Ans (9)**

$$\sigma^2 = \frac{20 \times 1 + 40 \times 2^2}{60} = 3 \quad (\text{Because mean of both samples is the same})$$

4. **Ans (4)**

$$5 \sin x = \ell(\sin x - 2 \cos x) + m(\cos x + 2 \sin x),$$

$$\ell + 2m = 5, m - 2\ell = 0$$

$$\therefore \ell = 1, m = 2$$

$$\begin{aligned} \text{So, } &\int \frac{5 \sin x}{\sin x - 2 \cos x} dx \\ &= \int 1 \, dx + 2 \int \frac{\cos x + 2 \sin x}{\sin x - 2 \cos x} dx = x + 2 \end{aligned}$$

$$\ln |\sin x - 2 \cos x| + c$$

$$A = 1, B = 2, 2A + B = 4$$

5. **Ans (2)**

$$\frac{dy}{dx} = \frac{y(1+xy)}{x} = \frac{y}{x} + y^2 \Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{xy} = 1$$

$$\text{Put } \frac{1}{y} = t, -\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}, -\frac{dt}{dx} - \frac{1}{x}t = 1$$

$$\frac{dt}{dx} + \frac{t}{x} = -1$$

If

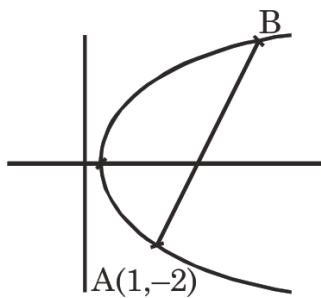
$$= e^{\int \frac{1}{x} dx} = x, t \cdot x = \int -1 \cdot x dx, \frac{x}{y} = -\frac{x^2}{2} + c$$

$$\frac{4}{2} = -8 + c \Rightarrow c = 10$$

$$\frac{x}{y} = 10 - \frac{x^2}{2} \Rightarrow y = \frac{2x}{20 - x^2}$$

$$f(4) = \frac{8}{20 - 16} = 2$$

6. **Ans (4)**



$$\text{Parameter of A : } 2 \times 1 \times t = -2$$

$$\therefore \text{Parameter of B : } t^2 = -(-1) - \frac{2}{-1} = 3$$

$$\therefore B(9, 6)$$

$$\therefore AB = 8\sqrt{2}$$

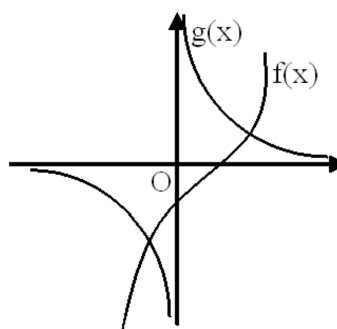
7. **Ans (2)**

$$e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$$

$$\Rightarrow (e^{3x} - 1)^2 - e^x (e^{3x} - 1) = 12e^{2x}$$

$$(e^{3x} - 1)^2 (e^x - e^{-x} - e^{-2x}) = 12$$

$$\Rightarrow \underbrace{e^x - e^{-x} - e^{-2x}}_{\text{increasing (let } f(x))} = \underbrace{\frac{12}{e^{3x} - 1}}_{\text{decreasing (let } g(x))}$$



$\Rightarrow$ No. of real roots = 2

8. **Ans (14)**

$$f(5) = 6, f(6) = 5 \rightarrow 2$$

$$f(5) = 6, f(6) \neq 5 \rightarrow 4$$

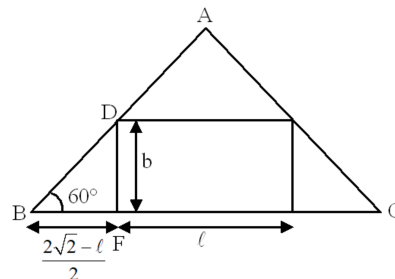
$$f(5) \neq 6, f(6) = 5 \rightarrow 2$$

$$f(5) \neq 6, f(6) \neq 5 \rightarrow 4$$

9. **Ans (6)**

$$4! \cdot (2! \cdot 3 + 2 \times 2 + 3) = 312$$

10. **Ans (3)**



In $\triangle DBF$

$$\tan 60^\circ = \frac{2b}{2\sqrt{2} - l} \Rightarrow b = \frac{\sqrt{3}(2\sqrt{2} - l)}{2}$$

A = Area of rectangle = $l \times b$

$$A = l \frac{\sqrt{3}}{2} (2\sqrt{2} - l)$$

$$\frac{dA}{dl} = \frac{\sqrt{3}}{2} (2\sqrt{2} - l) - \frac{l \cdot \sqrt{3}}{2} = 0$$

$$\boxed{l = \sqrt{2}}$$

$$A = l \times b = \sqrt{2} \times \frac{\sqrt{3}}{2} (\sqrt{2}) = \sqrt{3}$$

$$\Rightarrow \boxed{A^2 = 3}$$